L-08

Taylor and Maclaurin Polynomials and Series

Taylor polynomial

If f can be differentiated n times at x_0, then we define nth Taylor polynomial for f about $x=x_0$ to be,

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

$$= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k$$

Here,

$P_0 = f(x_0)$, $P_0(x) = f(x_0)$

$P_1(x) = f(x_0) + f'(x_0)(x-x_0)$

$P_2(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2$

$P_3(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3$

$\vdots$

Taylor series

If f has derivatives of all order at x_0, then we call the ~~series~~ Taylor series is,

$$f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2+\frac{f'''(x_0)}{3!}(x-x_0)^3+\cdots\cdots$$

[Infinite Taylor Series]

Ex 01

Find the Taylor series of the function $f(x)=\cos x$ at $x_0=\frac{\pi}{2}$. and also find P_0, P_1, P_2, P_3, P_4 polynomials.

Solution: Given, $f(x)=\cos x$ $\qquad f(\frac{\pi}{2})=\cos\frac{\pi}{2}=0$

$f'(x)=-\sin x \qquad f'(\frac{\pi}{2})=-\sin\frac{\pi}{2}=-1$

$f''(x)=-\cos x \qquad f''(\frac{\pi}{2})=-\cos\frac{\pi}{2}=0$

$f'''(x)=\sin x \qquad f'''(\frac{\pi}{2})=\sin\frac{\pi}{2}=1$

$f^{IV}(x)=\cos x \qquad f^{IV}(\frac{\pi}{2})=\cos\frac{\pi}{2}=0$

Now The polynomials

$P_0(x) = f(x_0)$

$= f(\frac{\pi}{2})$

$= 0$

$P_1 = f(x_0) + f'(x_0)(x - x_0)$

$= f(\frac{\pi}{2}) + f'(\frac{\pi}{2})(x - \frac{\pi}{2})$

$\therefore P_1(x) = 0 - (x - \frac{\pi}{2}) = -(x - \frac{\pi}{2})$

$P_2 = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2$

$= f(\frac{\pi}{2}) + f'(\frac{\pi}{2})(x - \frac{\pi}{2}) + \frac{f''(\pi/2)}{2!}(x - \frac{\pi}{2})^2$

$= 0 - (x - \frac{\pi}{2}) + 0$

$= -(x - \frac{\pi}{2})$

$P_3(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3$

$= f(\frac{\pi}{2}) + f'(\frac{\pi}{2})(x - \frac{\pi}{2}) + \frac{f''(\pi/2)}{2!}(x - \frac{\pi}{2})^2 + \frac{f'''(\pi/2)}{3!}(x - \pi/2)^3$

$= 0 - (x - \frac{\pi}{2}) + 0 + \frac{(x - \frac{\pi}{2})^3}{3!}$

$= -(x - \frac{\pi}{2}) + \frac{(x - \frac{\pi}{2})^3}{3!}$

$$P_4(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \frac{f^{IV}(x_0)}{4!}(x-x_0)^4$$

$$= f(\tfrac{\pi}{2}) + f'(\tfrac{\pi}{2})(x-\tfrac{\pi}{2}) + \frac{f''(\pi/2)}{2!}(x-\pi/2)^2 + \frac{f'''(\pi/2)}{3!}(x-\pi/2)^3 + \frac{f^{IV}(\pi/2)}{4!}(x-\pi/2)^4$$

$$= 0 - (x-\pi/2) + 0 + \frac{(x-\pi/2)^3}{3!} + 0$$

$$= -(x-\pi/2) + \frac{(x-\pi/2)^3}{3!}$$

<u>~~The~~ Taylor series</u>

$$f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \frac{f^{IV}(x_0)}{4!}(x-x_0)^4 + \cdots$$

$$= f(\pi/2) + f'(\pi/2)(x-\pi/2) + \frac{f''(\pi/2)}{2!}(x-\pi/2)^2 + \frac{f'''(\pi/2)}{3!}(x-\pi/2)^3 + \frac{f^{IV}(\pi/2)}{4!}(x-\pi/2)^4 + \cdots$$

$$= 0 - (x-\pi/2) + 0 + \frac{(x-\pi/2)^3}{3!} + 0 + \cdots$$

$$= -(x-\tfrac{\pi}{2}) + \frac{(x-\pi/2)^3}{3!} + \cdots$$

CamScanner

<u>Ex02</u> $f(x) = 2x^3 + 7x^2 - x - 1$

Expand the function $f(x)$ in powers of $(x-2)$ in infinite Taylor's series.

<u>Solution:</u> Given $f(x) = 2x^3 + 7x^2 - x - 1$

Here, $x_0 = 2$, $f(2) = 2(2)^3 + 7(2)^2 - 2 - 1$
$= 41$

~~Solution:~~

$f'(x) = 6x^2 + 14x - 1$, $f'(2) = 6(2)^2 + 14(2) - 1$
$= 51$

$f''(x) = 12x + 14$, $f''(2) = 12 \times 2 + 14$
$= 38$

$f'''(x) = 12$, $f'''(2) = 12$

<u>Now, the taylor series is</u>

$$f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \cdots$$

$$= f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3 + \cdots$$

$$= 41 + 51(x-2) + \frac{38}{2}(x-2)^2 + \frac{12}{6}(x-2)^3 + \cdots$$

$$= 41 + 51(x-2) + 19(x-2)^2 + 2(x-2)^3 + \cdots$$

(Ans.)

Ex 03

Expand $y=e^{ax}$ in the powers of $(x-1)$.

Solution: Given, $y=e^{ax}$

Here, $x_0=1$

$\therefore f(x)=e^{ax}$

$\therefore f(1)=e^{a.1}=e^{a}$

$f'(x)=ae^{ax}$ $\quad f'(1)=ae^{a.1}=ae^{a}$

$f''(x)=a^2e^{ax}$ $\quad f''(1)=a^2e^{a.1}=a^2e^{a}$

$f'''(x)=a^3e^{ax}$ $\quad f'''(1)=a^3e^{a.1}=a^3e^{a}$

Now the taylor series:

$$f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2!}(x-x_0)^2+\frac{f'''(x_0)}{3!}(x-x_0)^3+\cdots$$

$$=f(1)+f'(1)(x-1)+\frac{f''(1)}{2!}(x-1)^2+\frac{f'''(1)}{3!}(x-1)^3+\cdots$$

$$=e^a+ae^a(x-1)+\frac{a^2e^a}{2!}(x-1)^2+\frac{a^3e^a}{3!}(x-1)^3+\cdots$$

H.W (Ans)

1| Expand the function $f(x)=\cos x$ and $f(x)=\sin x$ in powers of $(x-\frac{\pi}{2})$ in infinite Taylor series.

2| Expand $\frac{1}{x+2}$ at $x_0=4$

3| Practice sheet

4| Anton Book Chapter 9.7 ⇒ (17-24) [Exercise]
Chapter 9.8 ⇒ (11-18)

Maclaurin polynomial

If ~~a~~ f can be differentiated n times at 0, then we define the nth Maclaurin polynomial for f to be,

$$P_n(x) = f(0) + f'(0)\,x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots + \frac{f^n(0)}{n!}x^n$$

$\boxed{\text{Here, } x_0 = 0}$

$$P_0(x) = f(0) \quad , \quad P_1(x) = f(0) + f'(0)\,x$$

$$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2$$

$$P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^2$$

Maclaurin Series

$$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

↳ Infinite series

Ex 04

Find the Maclaurin polynomial P_0, P_1, P_2, P_3 for $e^x \cos x$.

Solution: Given, $f(x) = e^x \cos x$

$f(0) = e^0 \cos 0$
$= 1.1$
$= 1$

$f'(x) = e^x \cos x - e^x \sin x = e^x(\cos x - \sin x)$

$f''(x) = e^x(\cos x - \sin x) + e^x(-\sin x - \cos x)$

$= e^x(\cos x - \sin x) - e^x(\sin x + \cos x)$

$= e^x(\cos x - \sin x - \sin x - \cos x)$

$= -2e^x \sin x$

$f'''(x) = -2e^x \sin x - 2e^x \cos x$

$= -2e^x(\sin x + \cos x)$

$f'(0) = e^0(\cos 0 - \sin 0) = 1(1-0) = 1$

$f''(0) = -2e^0 \sin 0 = 0$

$f'''(0) = -2e^0(\sin 0 + \cos 0)$

$= -2(1)(0+1)$

$= -2$

Now

Polynomials

$P_0(x) = f(0)$
$= 1$

$P_1(x) = f(0) + f'(0)x$
$= 1 + x$

$P_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2$

$= 1 + x + 0$
$= 1 + x$

$P_3(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$

$= 1 + x + 0 - \frac{2}{3!}x^3$ $[3! = 6]$
$= 1 + x - \frac{x^3}{3}$

(Ans)

Ex 05

Expand $y = \ln(x+1)$ in the power of x

Or, Find the Maclaurin series for $y = \ln(x+1)$ at $x_0 = 0$.

Or, Find the Maclaurin series for $y = \ln(x+1)$.

Solution Given, $y = \ln(x+1)$

$\therefore f(x) = \ln(x+1)$

$f(0) = \ln(0+1) = \ln(1) = 0$

$f'(x) = \frac{1}{x+1} = (x+1)^{-1}$

$f'(0) = \frac{1}{0+1} = 1$

$f''(x) = -1(x+1)^{-1-1} = -(x+1)^{-2}$

$= \frac{-1}{(x+1)^2}$

$= -(x+1)^{-2}$

$f''(0) = \frac{-1}{(0+1)^2} = -1$

$f'''(x) = 2(x+1)^{-2-1} = 2(x+1)^{-3}$

$= \frac{2}{(x+1)^3}$

$f'''(0) = \frac{2}{(0+1)^3} = 2$

[$2! = 2$]

[$3! = 3\times2\times1 = 6$]

The Maclaurin series,

$$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$$= 0 + x - \frac{1}{2!}x^2 + \frac{2}{3!}x^3 + \cdots$$

$$= x - \frac{x^2}{2} + \frac{2}{6}x^3 + \cdots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} + \cdots\cdots$$

Ex 06

Find the Maclaurin series for $\frac{1}{1-2x}$

Solution: Given, $f(x) = \frac{1}{1-2x} = (1-2x)^{-1}$

$$f(0) = \frac{1}{1-2(0)} = 1$$

$$f'(x) = -(1-2x)^{-1-1}(-2) = -(1-2x)^{-2}(-2)$$
$$= 2(1-2x)^{-2}$$
$$= \frac{2}{(1-2x)^2}$$

$$; f'(0) = \frac{2}{(1-2\times 0)^2} = 2$$

$$f''(x) = -4(1-2x)^{-2-1}(-2) = 8(1-2x)^{-3}$$
$$= \frac{8}{(1-2x)^3}$$

$$; f''(0) = \frac{8}{(1-2(0))^3} = 8$$

$$f'''(x) = -24(1-2x)^{-3-1}(-2) = 48(1-2x)^{-4}$$
$$= \frac{48}{(1-2x)^4}$$

$$; f'''(0) = \frac{48}{(1-2(0))^4} = 48$$

The Maclaurin series:

$$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$
$$= 1 + 2x + \frac{8}{2}x^2 + \frac{48}{6}x^3 + \cdots$$
$$= 1 + 2x + 4x^2 + 8x^3 + \cdots$$

(Ans)

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H.W

1) $y=\frac{\sin x}{\cos x}$. Expand the function in the power of x.

2) Expand the function $f(x)=\ln\left(\frac{1+x}{1-x}\right)$ in infinite Maclaurin series.

3) Expand the function $f(x)=e^{\cos x}$ at $x_0=0$.

4) Practice sheet

5) Anton Book ⇒ Chapter 9.7 ⇒ (7-12) [Exercise]
Chapter 9.8 ⇒ (~~10~~ 1-6)